

Chapter 1, Problem 32CP

Problem

A telephone wire has a current of $20\mu\text{A}$ flowing through it. How long does it take for a charge of 15 C to pass through the wire?

Step-by-step solution

Step 1 of 1

$$\begin{aligned} Q &= It \Rightarrow t = \frac{Q}{I} \\ &= \frac{15}{20 \times 10^{-6}} = 0.75 \times 10^6 \text{ sec} \end{aligned}$$